Pair-Native Post-TT-star Registry and Architecture-Reroute Interface

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Abstract

We freeze the exact finite interface for a pair-native route beginning after the TPC-18 Cauchy/TT-star step. The ordered bilinear row-pair carrier is source-backed, and the TPC-93 source–child inverse is exact once a retained source atom is supplied; its coefficient reassembly retains the physical squarefree and target-primitive support hypotheses. However, the declared TPC-205 registry/source-lock corpus contains no production pair record and no theorem-backed pair-to-source-atom crosswalk. Moreover, a TT-star bilinear pair term is not the linear, coefficient-wise conservative cut-to-occurrence edge required by H1. We give a 42-field fail-closed registry contract, two strictly finite non-production fixtures, and a complete normalization/loss firewall. The pair-native route therefore survives only as an open architecture-reroute candidate. This is an L0/L1 interface theorem, not a structural reopen, an L2 estimate, or a prime-pair theorem.

1 Authorization and level boundary

The authorized object is the finite interface

`FINITE_PAIR_NATIVE_POST_TTSTAR_REGISTRY_AND_`
`ARCHITECTURE_REROUTE_INTERFACE`.

Authorization is workflow input and not theorem evidence. The source snapshot is commit `88ead9383270ab40a6350616554d519ede6f0782`. At this snapshot TPC-204 has no complete direct-production crosswalk [7]. We use $L0$ for exact finite identities, schemas, and fixtures; $L1$ for source-locked formula and type assertions; and $L2 = \texttt{NONE}$.

2 The ordered post-TT-star carrier

Fix one TPC-18 opened-D packet. Write $\alpha = (\ell_\alpha, d_\alpha)$, $\gamma = (\ell_\gamma, d_\gamma)$, and $N_\alpha(j) = \ell_\alpha d_\alpha j + h_0$. TPC-18 gives [6]

$$|\mathcal{T}_D|^2 \ll_W J(\mathcal{E}_D + \mathcal{C}_D^{\mathrm{off}}), \tag{1}$$

where $\mathcal{C}_D^{\mathrm{off}}$ is summed over ordered triples (α, γ, j) in the same packet, with $\alpha \neq \gamma$. Its displayed coefficient carrier is

$$\mu(d_\alpha)\mu(d_\gamma)(\log \ell_\alpha)(\log \ell_\gamma) r_R(N_\alpha(j)) r_R(N_\gamma(j)) B_\alpha(j) B_\gamma(j). \tag{2}$$

Theorem 1 (Exact carrier and its boundary)**.** *The ordered summation domain in* (2)*, and the eight displayed coefficient factors, are source-backed. The second row γ is generated by the TT-star square expansion, not by a singleton cut. The diagonal is the separate energy $\mathcal{E}_D$. The factors B_α, B_γ remain source aliases whose full literal expansion is unsupplied; membership in the formal summation domain does not prove an active or nonzero occurrence.*

Proof. These statements are the literal domain, coefficient, and diagonal split of the TPC-18 opened-D formula. Its support restrictions are stated as "understood," so replacing the two B-aliases by products of archived single-row ASTs would add data not present in the source. □

TPC-32 supplies a separate matched-shell parent [9]:

$$w^{\mathrm{sh}}_{\alpha,\gamma,j} = \gamma^{(1)}_\alpha \gamma^{(2)}_\gamma \mathfrak{A}_{\alpha,\gamma}(j)\mathcal{K}^{\mathrm{sh}}_{\alpha,\gamma}(j), \tag{3}$$

$$\mathcal{K}^{\mathrm{sh}}_{\alpha,\gamma}(j) = C_{m_\alpha}(j)\mathsf{H}_{m_\gamma;T,U_0}(j) + \mathsf{H}_{m_\alpha;T,U_0}(j)C_{m_\gamma}(j). \tag{4}$$

Opening C_{m_α} produces $T < u \le U_0,\ u \mid N_\alpha(j)$; the right polarization is symmetric. Thus u is not a field of the parent (α, γ, j). TPC-32 also states that no complex conjugation is implicit. A Hermitian rewrite would therefore have to record the second-side conjugation explicitly. Nothing in these formulae identifies the TPC-18 B-carrier with the TPC-32 literal parent.

3 Conditional source–child composition

Given a retained TPC-93 source atom

$$\omega = (L/R, \alpha, \gamma, j, u),$$

and a divisor $v \mid (d_\alpha, d_\gamma)$, TPC-93 constructs a unique child (θ, t) and gives an explicit inverse back to ω [10]. Its projector identity is

$$\sum_{v|d,e} \lambda_{G^{\mathrm{row}}_X}(v) = \mathbf{1}_{(d,e)\le G^{\mathrm{row}}_X}. \tag{5}$$

A single child does not restore the source. The projector identity is a finite identity. On TPC-93's physical squarefree and target-primitive support, each child retains its sign and coefficient, and the weighted sum in (5) restores the actual-row-gcd-masked source coefficient. Both polarizations occur once and no new fiber normalization is introduced.

Theorem 2 (Conditional pair-native composition)**.** *Suppose a future source-backed record supplies a fully materialized ordered pair, the actual joint mask and packet schedule, T, U_0, a polarization, all normalization stages, and a theorem identifying that record with a retained ω on TPC-93's physical squarefree and target-primitive support. Then the TPC-93 source–child inverse and projector-weighted coefficient reassembly apply exactly. In the declared TPC-205 corpus the premise is not instantiated: the production pair-to-ω crosswalk is absent.*

Proof. The positive conclusion is precisely the TPC-93 inverse and (5), evaluated on the supplied source. Neither theorem constructs ω from a TPC-18 pair. Hence the stated premise cannot be deleted or supplied by renaming indices. □

4 Registry contract and finite nonvacuity

The machine contract requires 42 fields in four blocks:

Block	Count	Required separation
packet/identity scope	16	X, h_0, δ, scales, packet and source IDs
ordered pair	10	rows, j, targets, mask, coefficient AST, support/nonzero
source/child	12	polarization, u, σ, v, θ, t, inverse and resolved child
normalization	4	source, linear, quadratic, and target-return stages

Pairs are ordered and no swap quotient is allowed. `edge_instance_id` is distinct from `target_occurrence_id`, so a future theorem may represent legitimate many-to-one reassembly without collapsing transformation rows.

The first fixture is selected by semantic IDs from the TPC-133/TPC-136 archives [1, 2]:

$$((103, 1), (107, 1), 5), \qquad h_0 = 2.$$

It has

$$N_\alpha = 517, \quad N_\gamma = 537, \quad (N_\alpha, N_\gamma) = 1, \quad \Delta^\# = -4, \quad \det_{\mathrm{ord}} = -8.$$

Both cut records are FRONTIER_UNMAPPED/NO_TAIL_ROOM. Its only licensed label is DUAL_SOURCE_LOCKED_ROW_PAIR_CANDIDATE; the joint mask, pair ID, coefficient nonzero status, and production occurrence are absent.

For finite algebraic nonvacuity, the committed TPC-32 primitive fixture has

$$L = 100,\ R = 12,\ T = 50,\ U_0 = 200,\ h_0 = 2,\ j = 1, \quad \alpha = (59, 1),\ \gamma = (71, 1).$$

Feeding $u = 61$ into the TPC-93 formula gives

$$\sigma = v = 1, \quad d_0 = 0, \quad t = 1, \quad u_0 = 2, \qquad D(1) = 1, \quad U(1) = 61,$$

and determinant 2. These child fields are derived, not directly certified by TPC-32. The exact label is

TPC32_PRIMITIVE_FIXTURE_PLUS_TPC93_FORMULAS_DERIVED_LO_ONLY.

The integer $\sigma = 1$ here is not an analytic saving exponent.

5 Normalization and physical-loss firewall

The source-backed TPC-18 object in (1) is unnormalized. The archived string nu_X is only a scope label and not a scalar definition. If a future theorem supplies a multiplicative scalar c_X, then and only then one may write

$$|c_X \mathcal{T}_D|^2 \ll_W |c_X|^2 J(\mathcal{E}_D + \mathcal{C}_D^{\mathrm{off}}). \tag{6}$$

The square in $|c_X|^2$ and the four normalization stages are mandatory.

The ledger retains the source-backed prime-power error, bounded-overlap dyadic partition, Cauchy factor J, diagonal $\mathcal{E}_D \ll X^{1+\varepsilon}$, and the same/near/large-gcd bounds from TPC-18. It separately retains the TPC-25 zero, principal, drift and polylogarithmic terms [8], the TPC-32 drift and large-content bounds, and the TPC-93 Fourier tail. Three terminal rows remain:

$$\begin{aligned} \text{generic remainder} &= \texttt{UNCONTROLLED\_HARD\_REMAINDER}, \\ \text{square-root return} &= \texttt{MISSING\_UNSUPPLIED}, \\ \text{full-block/endpoint return} &= \texttt{MISSING\_UNSUPPLIED}. \end{aligned}$$

Every retained bound remains conditional on its own source theorem's hypotheses; the ledger does not compose these rows onto a production TPC-18 pair. Consequently no target positive exponent, strict 1/400 credit, or L2 credit is available.

6 H1 type separation and final verdict

The H1 occurrence lift is linear and coefficientwise conservative [3]:

$$L_X : \mathbb{C}^{\mathcal{C}_X^{\mathrm{ns}}} \longrightarrow \mathbb{C}^{\mathcal{O}_X}, \qquad \mathbf{1}_{\mathcal{O}_X}^T L_X = \mathbf{1}_{\mathcal{C}_X^{\mathrm{ns}}}^T. \tag{7}$$

Its conceptual entries may be signed or complex. The finite TPC-174 contract further uses nonzero exact-rational weights with per-cut column sum one, but its committed fixture is synthetic [4]. By contrast, the TPC-18 object is a TTSTAR_BILINEAR_PAIR_TERM generated after a quadratic inequality. There is no theorem inverse-aggregating it to every original cut coefficient.

Theorem 3 (Typed interface and first missing). *In the declared source-locked TPC-205 registry corpus, the pair and H1 relation types are distinct, the number of production pair records is zero, and the first production missing node is*

SOURCE_ LOCKED_ POST_ TTSTAR_ ORDERED_ PAIR_ REGISTRY_ WITH_ COMPLETE_ PAIR_ COEFFICIENT_ AND_ GLOBAL_ NORMALIZATION.

Thus H1.E is not repaired. The pair-native formula gate passes, while its production and structural reopen triggers fail. The pair-native architecture-reroute candidate remains open.

Proof. Theorems 1 and 2 establish the formula carrier and the conditional inverse. The declared source-locked registry has no row containing a joint mask, full literal coefficient, pair nonzero status, pair-to-ω theorem, and complete normalization return. Equation (7) has a different source type and no inverse-aggregation bridge. The conclusion follows without asserting a global impossibility theorem. □

TPC-179's independent roots for local edges E, actual active support A, and canonical/minimal representation M remain NOT TESTABLE [5]. Both O161 pointwise parents, H1, and the global architecture remain open. The scoped cell

`TPC18_TPC93_POST_TTSTAR_PAIR_DIRECT_COMPOSITION_V1=STOP_SCOPED`

stops only direct renaming of the current formula carrier as a production source atom or H1 edge; it does not stop a new registry, theorem crosswalk, or architecture.

7 Machine certificate and trust boundary

The active certificate locks 17 sources, four relation types, 42 required registry fields, two finite fixtures, 17 loss rows, and 23 gates. A separate L0 schema accepts only `ROW_ONLY` and `DERIVED_L0_ONLY`, never a production mode. The independent checker imports neither builder nor materializer; it reselects the four archive rows by semantic ID, recomputes their integrity hashes and integer arithmetic, reconstructs the derived affine child, freezes the complete nested contract by deep strict-type comparison, and independently executes 12 active-schema, 37 regenerated-schema semantic, and six Boolean/integer-confusion mutations. The manifest is a repository review pin, not an external signature or theorem source.

References

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